

Data Pricing via Competitive Equilibrium

Bhaskar Ray Chaudhury
University of Illinois
Urbana-Champaign, IL, USA
braycha@illinois.edu

Aniket Murhekar
Northwestern University
Evanston, IL, USA
aniket2@illinois.edu

Jugal Garg
University of Illinois
Urbana-Champaign, IL, USA
jugal@illinois.edu

Jiixin Song
University of Illinois
Urbana-Champaign, IL, USA
jiaxins8@illinois.edu

Abstract

Data powers almost everything we experience on the web today—from the recommendations and ads we see to the AI systems and online marketplaces that shape our digital interactions. The increasing demand for high-quality data has given rise to platforms that facilitate the buying and selling of data. A key practical challenge in such markets is determining how to price data. *Competitive equilibrium* (CE), a foundational concept in classical market economics, determines prices for *rivalrous* goods by matching their supply and demand. In this work, we initiate the study of CE in data markets, explicitly incorporating the role of data in improving predictive performance in buyers' utility functions, and the *non-rival* nature of data by adapting the standard market-clearing condition to allow the simultaneous allocation of data records to multiple buyers.

We analyze the *existence*, *structure*, and *computation* of CE in such data markets. We establish that CE always exists, and almost all instances admit a *unique* and *rational* equilibrium price vector. In general, however, there could be a non-convex set of prices, which rules out convex-programming approaches for finding a CE. Despite these challenges, we design an FPTAS for computing approximate equilibria using a Walrasian-style price adjustment algorithm. Our framework opens avenues for studying richer buyer utilities under correlated data sellers, and deeper structural and algorithmic aspects of data markets.

CCS Concepts

• **Theory of computation** → **Market equilibria; Computational pricing and auctions.**

Keywords

Data market, competitive equilibrium, data pricing

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1 Introduction

The rapid expansion of data-driven decision-making and the widespread deployment of data-centric technologies have firmly established data as one of the most valuable assets of the 21st century. Declining storage costs, coupled with major advances in data mining, analytics, and machine learning, have significantly increased both the utility and economic value of data. According to Acumen Research [48], the U.S. big data market is projected to reach approximately \$473 billion by 2030, underscoring the accelerating impact of data on innovation, productivity, and market dynamics.

A fundamental question in today's data-driven economy is: *how should data be priced?* In this paper, we initiate a formal study of the competitive pricing of data. Classical economic theory discusses two forms of market competition—*oligopolistic* and *perfect competition* [42]. In oligopolistic markets, a small number of sellers possess sufficient market power to influence prices. In contrast, in perfectly competitive markets, all sellers act as *price-takers*, responding to prices set by a *Walrasian auctioneer* who ensures that supply equals demand for every good. Our focus in this paper is on perfectly competitive environments, and we adopt *competitive equilibrium* (CE) [7] as our solution concept. We establish positive results on the existence, uniqueness, and computability of stable outcomes (CE) in perfectly competitive markets.

While CE has been extensively studied – both structurally and algorithmically – in traditional markets, its application to data introduces distinct challenges and structural subtleties. Formally, we study a data economy consisting of a set N of n budget-constrained buyers interested in acquiring data, and a set M of m sellers, each endowed with a collection of data records offered for sale. These records could include, for example, in-store purchase histories of a retailer, time-series data tracking an asset's price, or transaction logs from a credit card company. Our goal is to determine prices for these data endowments such that, when buyers choose data optimally according to their preferences and budgets, the overall demand for each seller's data matches its supply, i.e., all available data is fully utilized.

Value of data. To define optimal data bundles for buyers, it is essential to understand how buyers derive utility from data. Consistent with Shannon's information-theoretic perspective, we model data as an asset that reduces uncertainty. We focus on settings where buyers are typically machine learning (ML) agents aiming to improve the accuracy of their predictions. In such contexts, more training data helps improve the precision of these predictions.

Each buyer $i \in N$ seeks to estimate an unknown parameter θ_i , which represents a latent quantity of interest such as a future market signal, risk score, or asset price. Each data record is interpreted as a digitized signal providing noisy information about θ_i . Signals originating from a single seller follow a fixed prior distribution, depending on the seller's specific economic activity and the relevance of the data generated to predict θ_i . The signal distributions across sellers vary due to heterogeneity in data generation processes and their relevance. For example, consider a buyer aiming to forecast the future price of a financial asset (θ_i). Data from a brokerage platform may include investor behavior and transaction volumes, while another seller, such as a news analytics provider, offers sentiment scores extracted from financial news articles. Both data sources provide noisy signals about the same underlying parameter, but differ in structure and reliability.

A data bundle $\mathbf{x}_i = (x_{i,1}, x_{i,2}, \dots, x_{i,m})$ represents a collection of data records aggregated from multiple sellers, where $x_{i,j}$ denotes the quantity of data records acquired by buyer i from seller j . Upon receiving the full set of signals $S(\mathbf{x}_i)$ – one signal per data record – the buyer updates her belief about the latent variable θ_i to $\theta_i | S(\mathbf{x}_i)$.

Consistent with the growing literature on the value of data [9], we define the buyer's *utility* from the data bundle, denoted by $u_i(\mathbf{x}_i)$, as the reduction in uncertainty about θ_i :

$$u_i(\mathbf{x}_i) = \mathbb{E} [\text{Pre}(\theta_i | S(\mathbf{x}_i))] - \text{Pre}(\theta_i),$$

where the precision of a random variable, $\text{Pre}(\theta)$, is defined as the inverse of its variance, i.e., $\text{Pre}(\theta) = 1/\text{Var}(\theta)$.

Market clearing for data. In a conventional market economy, a price vector $\mathbf{p} = (p_1, p_2, \dots, p_m)$, where p_j denotes the price per unit of good j , together with an allocation $\mathbf{x} = (x_1, x_2, \dots, x_n)$, is said to be a competitive equilibrium (CE) if two conditions hold: (i) each buyer i exhausts her budget b_i on an optimal bundle, i.e., $\mathbf{x}_i \in \arg \max_{\mathbf{y}} \{u_i(\mathbf{y}) : \langle \mathbf{p}, \mathbf{y} \rangle \leq b_i, \mathbf{y} \geq \mathbf{0}\}$, and (ii) the *market clears* each good, meaning that for each j the total demand ($\sum_i x_{i,j}$) equals the total supply s_j , i.e., $\sum_i x_{i,j} = s_j$ for all $j \in M$.

When extending this framework to data economies, the first condition extends naturally. However, a key distinction arises in enforcing the second condition, due to the *non-rivalrous* nature of data. Unlike traditional goods, data records can be duplicated at negligible cost and simultaneously consumed by multiple buyers. As a result, the total consumption of a seller's dataset may exceed the supply s_j . Nevertheless, the spirit of market clearing – that all available resources are fully utilized – remains meaningful. In the context of non-rivalrous goods such as data, this translates to requiring that each seller's dataset be fully demanded by at least one buyer, i.e., $\max_i x_{i,j} = s_j$. This form of market clearing for non-rivalrous assets has also found precedent in the context of digital goods [41]. However, while [41] primarily address the existence of competitive equilibria under general utility functions, our focus lies in the geometry, computation, and economic properties of equilibrium when utility functions have a more structured form – specifically, those that capture improvements in prediction accuracy, measured by the reduction in posterior variance relative to the prior.

Before presenting our main results, we introduce a simple illustrative example that highlights the distinction between rivalrous

and non-rivalrous economies and offers a welfare-related insight into CE in the rivalrous case.

Example 1. Consider two buyers with \$1 of budget each and one good $\{g\}$. Both buyers have strictly positive marginal utility from consuming g . If g is rivalrous, the market-clearing price is \$2, and each buyer receives 0.5 units of the good. In contrast, if g is non-rivalrous, the market-clearing price drops to \$1, and each buyer can consume one full unit of g . Observe that the total revenue earned by the seller in both settings is the same, but the buyers are strictly better off in the non-rivalrous setting (unsurprisingly). Moreover, in the non-rivalrous case, lowering the price of g reduces the seller's revenue without improving buyer utility, while raising the price diminishes buyer utility without increasing seller revenue. Therefore, the market-clearing price of \$1 is optimal for both the seller and the buyers.

In the example above, the utility profile under the non-rivalrous competitive equilibrium (CE) Pareto-dominates that of the rivalrous CE. This naturally raises the question of whether such Pareto-domination holds more generally, as non-rivalry allows for greater potential in consumption and, consequently, higher utilities. We answer this question affirmatively in Lemma 3.2. Further, in Section 3, we present additional instances that highlight more structural and economic differences between CEs in rivalrous and non-rivalrous settings.

1.1 Our Contributions

In this paper, we offer several results on the geometry and computational properties of a CE. Recall that in our setting a CE comprises of a price vector $\mathbf{p}$, and a data allocation $\mathbf{x}$, such that (i) every buyer i gets her optimal demand bundle, i.e., $\mathbf{x}_i \in \arg \max \{u_i(\mathbf{y}) : \langle \mathbf{p}, \mathbf{y} \rangle \leq b_i, \mathbf{y} \geq \mathbf{0}\}$, and (ii) for each seller j , demand equals supply, i.e., $\max_{i \in N} x_{i,j} = s_j$.

Existence and structure of CE. We show that a CE always exists when buyer utilities capture improvements in prediction accuracy (Theorem 4.1 and Remark 2). While the structure of the CE shares some similarities with that of classical markets (e.g., the Fisher market [23]), it also exhibits key differences. To state our results, we remark that each utility function $u_i(\cdot)$ is characterized by a set of parameters $(\tau_{i,j})_{j \in M}$, where each $\tau_{i,j}^{-1}$ is the variance of the noise in the signals generated by the data of seller j ; see Eq. (2) in Section 2 for the explicit form. In fact, one can show that $\mathbf{x}_i$ is an optimal bundle for buyer i if and only if (i) $\sum_j x_{i,j} p_j = b_i$, and (ii) $x_{i,j} > 0$ only if $\tau_{i,j}/p_j$ is maximum (this condition is referred to as the bang per buck condition). We characterize the set of *non-degenerate* data market instances as those where the $\tau_{i,j}$ s are not roots of a (special) multivariate polynomial in nm variables.

We prove that the competitive equilibrium (CE) price vector is unique for all non-degenerate instances, closely mirroring the behavior observed in Fisher markets. Uniqueness of prices is a highly desirable property, as it ensures predictability in the final market outcome, thereby offering clear advantages for data buyers, sellers, and policymakers.

THEOREM 1.1. *A CE always exists. For non-degenerate instances, there exists a unique, rational CE price vector.*

Algorithm. In many market settings, the uniqueness of competitive equilibrium (CE) is often indicative of an underlying convex

program whose optimal solutions correspond to CE outcomes (e.g., Eisenberg–Gale [31], Shmyrev [50], Eisenberg–Gale Dual [23]). However, our setting presents several challenges to formulating such a program. *Notably, we find that degenerate instances can admit multiple CE price vectors, and more surprisingly, the set of equilibria can be non-convex!* As a result, any convex programming formulation must be carefully tailored to non-degenerate instances. To the best of our knowledge, no existing convex program for CE in rivalrous goods markets explicitly leverages non-degeneracy in this way.

Additionally, in standard rivalrous goods economies, convex programs often exhibit a clean primal-dual structure: prices appear as primal variables, while allocations arise as dual variables (or vice versa). Typically, this duality ensures that the set of feasible allocations forms a convex cone at equilibrium prices. In contrast, our setting breaks this structure. We present non-degenerate instances where the set of market-clearing allocations corresponding to a fixed price vector is non-convex.

In summary, these observations point to several fundamental and intriguing challenges in designing continuous optimization frameworks for computing CE prices in data markets – challenges that we believe open up a rich direction for future research. Despite the foregoing challenges to developing a convex program to capture the set of CE, we successfully design a natural auction-style *Walrasian algorithm* that computes an arbitrary good approximation of CE in polynomial time. In an ϵ -approximate CE, every buyer receives a bundle that gives them at least a $(1 - \epsilon)$ fraction of their optimal utility, and each dataset is fully consumed.

THEOREM 1.2. *There exists a Walrasian algorithm that computes a ϵ -CE in data marketplaces in time polynomial in the input and $1/\epsilon$.*

The algorithm is iterative, and at its core follows a simple principle: “*Increase the prices of the datasets in demand*”. We now provide a short overview of our algorithm: we maintain prices $\mathbf{p}$, and allocation $\mathbf{x}$, that satisfy:

- (i) *Bang-per-buck spending*, where every agent purchases an optimal bundle of datasets, by spending a portion of their budget ($x_{i,j} > 0$ only if $\tau_{i,j}/p_j$ is maximum), and
- (ii) *Market-clearing*, i.e., every dataset has at least one agent that demands it completely ($\max_i x_{i,j} = 1$ for all j).

Note that if both the foregoing conditions are satisfied with *budget exhaustion*, i.e., every agent spends all their budget, then $(\mathbf{x}, \mathbf{p})$ forms a CE. The goal of our algorithm is to achieve budget exhaustion while preserving the above invariants. We begin by setting all dataset prices to very low values, ensuring that the initial allocations satisfy the invariants. As long as there are agents with sufficient excess budgets, we can sufficiently increase the prices of the datasets they are interested in (we refer to these datasets as “*datasets in demand*”), while still maintaining the invariants. Importantly, invariant (ii) ensures that the price of any dataset remains bounded above by the maximum budget of any agent. Consequently, our algorithm is guaranteed to terminate after a polynomial number of iterations.

We remark that while our algorithm exhausts all budgets eventually, it does not necessarily ensure monotonicity in budget exhaustion, i.e., the total excess budget across agents may not decrease in

every iteration. Nonetheless, the procedure converges to a state of budget exhaustion. Investigating algorithms and dynamics that not only achieve budget exhaustion but also guarantee a monotonic decrease in excess budgets is a natural and intriguing direction for future research.

1.2 Related Work

Economics of data, and data markets. With the proliferation of AI and data-centric technologies across industries, data economics has rapidly evolved into a vital area of study. A full survey is well beyond the scope of this paper, but we list the work that most closely resonates with our contributions. In this paper, we have adopted a very generic model for data valuations, defining an agent’s utility as an improvement in prediction accuracy. There have been more domain-specific studies on how agents value data [36, 37, 53]. We refer the reader to [38] for a detailed review of data valuation methods.

Given the significance of data to ML tasks, it is no surprise that there has been a significant body of work on data acquisition mechanisms that incentivize agents to share data [25, 34, 35, 44], by compensating them for their privacy loss. There have been studies on mechanisms that incentivize sellers to truthfully report the variances of their datasets to a data aggregator, who is tasked with attaining a certain variance level for her prediction [26].

Data-markets are two-sided marketplaces that receive prediction requests from buyers and address these requests from the datasets hosted on them by data sellers. Several questions on truthfulness, incentives, and revenue maximization have been studied in data markets. There is a line of work [2, 3, 8, 13] that investigates revenue-maximizing strategies of a monopolist data seller. [5] design principles and algorithms for a centralized online data marketplace where there are multiple buyers seeking data for their private ML tasks, and multiple sellers hosting their datasets on the platform. This model talks about pricing, training, revenue collection, and fair revenue distribution among the sellers. There is also a line of work that discusses equilibrium and auctions in data markets in the presence of externalities [4, 40]. Several studies investigate the pricing of data/ information from other first principles in different settings [12, 15, 43, 47]. In contrast to our monetary setting, prior work has investigated stable solutions in data exchange economies, where agents exchange data without monetary transfers [6, 14, 51].

Competitive equilibrium. The problem of computing CE in traditional rivalrous economies has been extensively studied. We focus here on the most relevant work for Fisher markets with linear utilities. In such markets, the CE set is captured by the Eisenberg–Gale convex program [31], which maximizes Nash social welfare—defined as the geometric mean of agents’ utilities. Shmyrev [50] later proposed an alternative convex formulation for this problem. Cole et al. [23] established duality connections between these and other related convex programs, and also provided a convex formulation for the case of quasi-linear utilities. Devanur et al. [28] gave a combinatorial polynomial-time algorithm for computing CE, which was subsequently improved to strongly polynomial-time algorithms by Orlin [46] and Végh [52]. Polynomial-time computability extends to more general classes such as homogeneous and weak gross substitutes (WGS) utility functions [10, 21, 30], but beyond these,

computing a CE becomes essentially PPAD-hard, particularly when the CE set is non-convex [17, 19, 20, 22, 27, 39, 49].

2 Model and Preliminaries

In this section, we introduce our model of a data marketplace, the notion of competitive pricing of data in this marketplace, and a characterization of competitive equilibria. Due to space constraints, some proofs are omitted from the main text; all proofs missing from the main body appear in the Appendix.

Data marketplace. A data market consists of a set $N = [n]$ of n buyers and a set $M = [m]$ of m sellers. Each seller j has an endowment of s_j data records, $\mathcal{D}_j$. Similar to the literature in [1, 11, 12, 26, 32, 33, 45, 54], we treat data production as a byproduct of economic activities and mining operations. The volume and quality of data available to a seller depend on the scale and nature of their economic operations, as well as the sophistication of their ML and data engineering capabilities. Sellers who are more economically active and technically proficient tend to have larger and richer data endowments. *Crucially, in what follows, we will treat the data endowment s_j of seller j as a hard supply constraint: it represents the maximum number of data records of seller j that can be used in the market at an instant in time.*

Each buyer i , on the other hand, aims to improve her estimate of an unknown state $\theta_i \sim N(\mu_i, \tau_i^{-1})$ with the help of the data available in the market. Based on the existing literature on data valuations [12, 32, 33], we interpret each datapoint from seller j , as an unbiased signal s_{ij} to buyer i about θ_i . In particular,

$$s_{ij} = \theta_i + \eta_{ij} \quad (1)$$

Here, the noise term, $\eta_{ij} \sim N(0, \tau_{ij}^{-1})$, follows a Gaussian distribution with zero mean and variance τ_{ij} . Intuitively, τ_{ij} captures the relevance of seller j 's endowment data-distribution to buyer i 's prediction task. Additionally, each buyer i is subject to a budget constraint $b_i > 0$, which limits the total amount she is willing to spend on acquiring data to improve her prediction.

Data allocation. A data allocation (or simply, allocation) is represented by a matrix $\mathbf{x} = (x_{ij})_{i \in N, j \in M}$, where $x_{ij} \in \mathbb{R}_{\geq 0}$ denotes the number of data records of seller j that buyer i receives, i.e., for each seller j , buyer i observes x_{ij} independent signals of the form given in Equation (1). We let $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,m})$ denote the allocation of data to agent i , and $S(\mathbf{x}_i)$ denote the corresponding set of signals.

Utility of data. We represent buyer preferences over data allocations using utility functions $u_i : \mathbb{R}_{\geq 0}^m \rightarrow \mathbb{R}_{\geq 0}$, where $u_i(\mathbf{x}_i)$ denotes the utility that buyer i derives from receiving data bundle $\mathbf{x}_i$. This utility reflects the improvement in buyer i 's precision when estimating θ_i , i.e., a reduction in her uncertainty following observation of the signals in $S(\mathbf{x}_i)$.

Formally, let $\text{Pre}(\theta_i)$ denote the prior precision of θ_i , defined as the inverse of its variance ($\text{Pre}(\theta) = 1/\text{Var}(\theta)$), and $\mathbb{E}[\text{Pre}(\theta_i | S(\mathbf{x}_i))]$ the precision of the posterior mean of θ_i after observing signals $S(\mathbf{x}_i)$. Define α_i as buyer i 's value per unit gain in the precision of θ_i , that is, her value for uncertainty reduction. Then, we define buyer i 's utility from bundle $\mathbf{x}_i$ as:

$$u_i(\mathbf{x}_i) = \alpha_i \cdot (\mathbb{E}[\text{Pre}(\theta_i | S(\mathbf{x}_i))] - \text{Pre}(\theta_i)) .$$

The term $\mathbb{E}[\text{Pre}(\theta_i | S(\mathbf{x}_i))]$ captures the gain in information from observing the signals $S(\mathbf{x}_i)$; more informative signals leads to a tighter distribution over the posterior expectation of θ_i , $\mathbb{E}[\theta_i | S(\mathbf{x}_i)]$. Given that prior (θ_i) and each of the signals in $S(\mathbf{x}_i)$ follow normal distribution, it is straightforward to show that the posterior ($\theta_i | S(\mathbf{x}_i)$) also follows a normal distribution, yielding the following closed-form expression for the utility functions:

$$u_i(\mathbf{x}_i) = \alpha_i \cdot \sum_{j=1}^m \tau_{i,j} \cdot x_{i,j} . \quad (2)$$

Furthermore, we make the natural assumption that each seller's data benefits at least one buyer, i.e., for every dataset j , we have $\tau_{i,j} > 0$ for at least one buyer i . For each $i \in N$, let τ_i denote the vector $\langle \tau_{i,1}, \dots, \tau_{i,m} \rangle$. Let $\tau_{\min} = \min(\min\{\tau_{i,j} | \tau_{i,j} > 0, i \in N, j \in M\}, \min\{\tau_i | i \in N\})$ and $\tau_{\max} = \max(\max_{i \in N} \tau_i, \max_{i,j} \tau_{i,j})$. Let $b_{\min} = \min_{i \in N} b_i$ and $b_{\max} = \max_{i \in N} b_i$.

Pricing data. We are interested assigning a price p_j to each dataset $\mathcal{D}_j$, which specifies the price for each data record purchased from seller j . Thus, if a buyer i is allocated $x_{i,j}$ data records of dataset $\mathcal{D}_j$, then she is obligated to pay $p_j \cdot x_{i,j}$ to seller j . At prices $\mathbf{p} = (p_j)_{j \in M}$, the spending of buyer i in allocation $\mathbf{x}$ is $\langle \mathbf{p}, \mathbf{x}_i \rangle := \sum_j p_j \cdot x_{i,j}$. An allocation $\mathbf{x}$ is feasible at the price vector $\mathbf{p}$ if no buyer spends more than her budget, i.e., $\langle \mathbf{p}, \mathbf{x}_i \rangle \leq b_i$ for all $i \in N$.

Competitive equilibrium. In traditional economies with rivalrous goods, a competitive equilibrium (CE) is a set of prices for the goods $\mathbf{p}$ and an allocation $\mathbf{x}$ such that (1) every buyer receives an optimal bundle of goods subject to the budget constraint, i.e., $\mathbf{x}_i \in \arg \max_{\mathbf{y}} \langle \mathbf{p}, \mathbf{y} \rangle \leq b_i, u_i(\mathbf{y})$, and (2) the total demand equals total supply, i.e., $\sum_i x_{ij} = s_j$ for each good j .

We observe that the second condition ($\sum_i x_{ij} = s_j$) is unsuitable in the context of non-rivalrous commodities like data. Since each data record can be duplicated at zero marginal cost and consumed by multiple agents, the total data records of $\mathcal{D}_j$ consumed in a data market can potentially exceed s_j . However, the core principle of market clearing – all produced goods are fully utilized – still applies. In the case of non-rivalrous goods such as data, this translates to requiring that each dataset $\mathcal{D}_j$ is fully demanded by at least one agent, i.e., for every $j \in M$, $\max_{i \in N} x_{ij} = s_j$.

With this definition of market clearing, the definition of CE in a data marketplace follows naturally. A price vector $\mathbf{p}$ along with a data allocation $\mathbf{x}$ is a CE if (1) every buyer receives the maximum utility that she can receive in any feasible data allocation at price vector $\mathbf{p}$, and (2) the market clears by ensuring that each dataset is fully consumed by some buyer. Formally, we introduce the following generalized definition of ε -approximate CE for algorithmic considerations, while noting that $\varepsilon = 0$ corresponds to exact CE.

Definition 2.1 (ε -competitive equilibrium). A price vector $\mathbf{p}$ and a feasible allocation $\mathbf{x}$ at $\mathbf{p}$ form an ε -competitive equilibrium (ε -CE) if they satisfy the following conditions:

- (i) (Approximately optimal bundle) Every buyer receives an approximately optimal bundle of datasets at prices $\mathbf{p}$ among all feasible allocations: for all $i \in N$, $u_i(\mathbf{x}) \geq (1 - \varepsilon) \cdot u_i(\mathbf{y})$ for all allocations $\mathbf{y}$ such that $\langle \mathbf{p}, \mathbf{y}_i \rangle \leq b_i$ and $\mathbf{y} \geq 0$.

- (ii) (Market-clearing) Every dataset is fully consumed by some buyer: for all $j \in M$, $\max_{i \in N} x_{i,j} = s_j$.

REMARK 1. Note that in any ε -CE $(\mathbf{p}, \mathbf{x})$ we have $p_j > 0$ for all $j \in M$, i.e., all prices are positive. For the sake of contradiction, suppose $p_j = 0$ for some $j \in M$. Then consider a buyer i such that $\tau_{i,j} > 0$; such a buyer exists by assumption. Since $p_j = 0$ and $\tau_{i,j} > 0$, the bundle $\mathbf{y}$ where $y_j \rightarrow \infty$ in the approximately optimum bundle for buyer i , and this violates market-clearing as $\max_i x_{i,j} \geq y_j \gg s_j$.

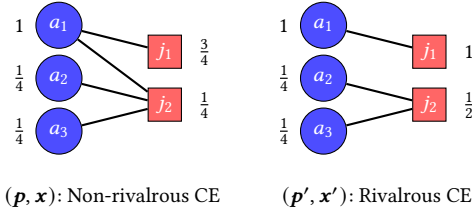
We introduce the structure of maximum bang-per-buck (MBB) graphs, which are useful for characterizing CE.

Definition 2.2 (ε -MBB-graph). Given a data marketplace and a parameter $\varepsilon > 0$, the ε -MBB-graph corresponding to a price vector $\mathbf{p}$ is defined as a bipartite graph $G_{\mathbf{p}}(\varepsilon) = (N \sqcup M, E_{\mathbf{p}}(\varepsilon))$, where the buyers and sellers are the two disjoint sets of vertices and $(i, j) \in E_{\mathbf{p}}(\varepsilon)$ if and only if $\tau_{i,j}/p_j \geq (1 - \varepsilon) \cdot \max_{j' \in M} \tau_{i,j'}/p_{j'}$.

Now, we present an example to illustrate a CE $(\mathbf{p}, \mathbf{x})$, and contrast it with the equilibrium $(\mathbf{p}', \mathbf{x}')$ that would emerge under the same conditions if the datasets were treated as rivalrous. Observe that both equilibria have distinct market-clearing prices, MBB structure, and each agent obtains strictly higher utility in $\mathbf{x}$ than in $\mathbf{x}'$.

Example 2. Consider an instance with three buyers a_1, a_2 , and a_3 with budgets $b_1 = 1$ and $b_2 = b_3 = \frac{1}{4}$ and two datasets j_1 and j_2 . The input parameters are $\tau_1 = (3, 1)$, and $\tau_2 = \tau_3 = (1.5, 1)$. Consider the prices $\mathbf{p} = (\frac{3}{4}, \frac{1}{4})$. Both datasets yield the same bang-per-buck for buyer 1, and only the second dataset provides the MBB for the other two buyers. Then the allocation $\mathbf{x} = (x_1, x_2, x_3)$ given by $x_1 = (1, 1)$, $x_2 = (0, 1)$, and $x_3 = (0, 1)$ forms a CE with $\mathbf{p}$.

In contrast, had the datasets been rivalrous, the CE $(\mathbf{p}', \mathbf{x}')$ is given by $\mathbf{p}' = (1, \frac{1}{2})$ and allocation bundles $\mathbf{x}'_1 = (1, 0)$, $\mathbf{x}'_2 = \mathbf{x}'_3 = (0, \frac{1}{2})$. The MBB graphs of the two equilibria are illustrated below.



We now identify some helpful sufficient conditions on $\mathbf{p}$ and $\mathbf{x}$ that guarantee that $(\mathbf{p}, \mathbf{x})$ is an approximate-CE.

LEMMA 2.3. Let $\varepsilon_1, \varepsilon_2 > 0$ be parameters. Suppose $(\mathbf{p}, \mathbf{x})$ satisfies the following conditions:

- (i) (Approximate budget exhaustion) For each buyer $i \in N$, $(1 - \varepsilon_1) \cdot b_i \leq \langle \mathbf{p}, \mathbf{x}_i \rangle \leq b_i$.
- (ii) (Approximately maximum bang-per-buck) For each buyer $i \in N$ and seller $j \in M$, $x_{i,j} > 0$ only if $(i, j) \in E_{\mathbf{p}}(\varepsilon_2)$.
- (iii) (Market-clearing) For each seller $j \in M$, $\max_{i \in N} x_{i,j} = s_j$.

Then $(\mathbf{p}, \mathbf{x})$ is an $(\varepsilon_1 + \varepsilon_2)$ -CE.

Normalization of supply. Note that our model does not place any restriction on the supply s_j of dataset $j \in M$. However, we show in Lemma B.1 that for computing (approximate) competitive

equilibria, it is without loss of generality to assume that $s_j = 1$ for all $j \in M$. With this normalization, $x_{i,j}$ represents the fraction of dataset j allocated to buyer i in allocation $\mathbf{x}$.

3 Structural Properties of Competitive Equilibrium

In this section, we investigate some basic structural properties of CE and contrast them with the structure of CE in the classic rivalrous goods economy. Our first observation is that for most instances, the CE prices are unique. Uniqueness is a very desirable feature of CE, as it is often considered a good indicator of the predictability and stability of a market.

Uniqueness of CE prices for non-degenerate instances. Aligning with established literature on competitive equilibrium, we define an instance as *non-degenerate* if the MBB graph $G_{\mathbf{p}}$ is acyclic for every price vector $\mathbf{p}$. It is well known that any instance can be approximated arbitrarily closely by a non-degenerate instance: In particular, for every instance $\mathcal{I}$, there exists a sequence of non-degenerate instances $\{\mathcal{I}_\ell\}_{\ell \geq 1}$ that converge to $\mathcal{I}$, i.e., $\lim_{\ell \rightarrow \infty} \mathcal{I}_\ell = \mathcal{I}$ [29]. We first observe that the CE price vector is unique for non-degenerate instances.

THEOREM 3.1. For any non-degenerate data market instance, there exists a unique CE price vector.

PROOF. Assume otherwise. Let $\mathbf{p}$ and $\mathbf{p}'$ be two CE price-vectors, and let $\mathbf{x}$ and $\mathbf{x}'$ be corresponding market clearing allocations. Without loss of generality, there exists at least one $j \in [m]$, such that $p'_j > p_j$ (otherwise we change the roles of $\mathbf{p}$ and $\mathbf{p}'$). Let $S \subseteq M$ be the datasets that have maximum multiplicative price increase from $\mathbf{p}$ to $\mathbf{p}'$, i.e., for each $j \in S$, $p'_j/p_j = \max_{\ell} p'_\ell/p_\ell$. We first observe that buyers that have MBB-edges to S in $G_{\mathbf{p}'} = (N, E_{\mathbf{p}'})$ can only have MBB edges to S in $G_{\mathbf{p}} = (N, E_{\mathbf{p}})$.

Claim 1. Suppose $(i, j) \in E_{\mathbf{p}'}$ where $j \in S$. Then $(i, j) \in E_{\mathbf{p}}$ and for all $j' \notin S$, we have $(i, j') \notin E_{\mathbf{p}}$.

PROOF. To prove the claim in the lemma, it suffices to argue that $\tau_{i\ell}/p_\ell \geq \tau_{i\ell}/p'_\ell$ for all ℓ , and the inequality is strict for all $\ell \notin S$. We prove this by contradiction. Assume otherwise, and let ℓ be such that $\tau_{i\ell}/p_\ell > \tau_{i\ell}/p'_\ell$, or $\ell \notin S$ and $\tau_{i\ell}/p_\ell \geq \tau_{i\ell}/p'_\ell$. By definition of S , we have $p'_\ell/p_\ell \leq p'_j/p_j$ (inequality being strict for all $\ell \notin S$), implying that $\tau_{i\ell}/p'_\ell > \tau_{i\ell}/p'_j$. This implies that $(i, j) \notin E_{\mathbf{p}'}$, which is a contradiction. $\square$

Let $T \subseteq N$ be the set of buyers that consume at least one dataset in S completely in the CE $(\mathbf{x}', \mathbf{p}')$, i.e., $T = \{i \in N \mid \exists j \in S, \text{ s.t. } x'_{i,j} = 1\}$. For every buyer $i \in T$, let $N'(i) = \{j \in S \mid x'_{i,j} = 1\}$. Since $\mathbf{x}'$ is a market clearing allocation, we have $\sum_{i \in T} |N'(i)| \geq |\cup_{i \in T} N'(i)| = |S|$. We now make a crucial observation: every $i \in T$ has at least $|N'(i)| + 1$ MBB edges to buyers in S in $G_{\mathbf{p}}$. First observe that every MBB-edge from T to S in $G_{\mathbf{p}'}$ is also an MBB edge from T to S in $G_{\mathbf{p}}$, by Claim 1. Now, note that $p'_j > p_j$ for all $j \in S$, implying that $b_i \geq \sum_{j \in N'(i)} p'_j > \sum_{j \in N'(i)} p_j$, further implying that i needs to have an MBB edge to datasets outside of $N'(i)$ to clear its budget in the CE $(\mathbf{p}, \mathbf{x})$. By Claim 1, this MBB edge should be to some buyer in S itself. Therefore, every buyer $i \in T$ has at

least $|N'(i)| + 1$ MBB-edges to datasets in S in $G_{\mathbf{p}}$. Thus, the induced subgraph $G_{\mathbf{p}}[T \cup S]$ has at least $\sum_{i \in T} (|N'(i)| + 1) \geq |S| + |T|$ many edges, implying that it is cyclic, which is a contradiction. $\square$

Improved utility guarantees compared to rivalrous CE. We compare the utilities of buyers in a CE as compared to their utilities from a CE of a rivalrous goods instance. Given an instance $\mathcal{I}$, let $(\mathbf{p}', \mathbf{x}')$ be the CE as per Definition 2.1, i.e., requiring $\max_j x_{i,j} = 1$ for all $j \in M$. Let $(\mathbf{p}, \mathbf{x})$ be the CE of the same instance by treating the goods as rivalrous, i.e., requiring $\sum_i x_{i,j} = 1$ for all $j \in M$.

LEMMA 3.2. *For every $i \in N$, $u_i(\mathbf{x}') \geq u_i(\mathbf{x})$. Thus, every agent receives as much utility in a CE with non-rivalrous datasets compared to a CE where the datasets are rivalrous.*

PROOF. We first prove that $p'_j \leq p_j$ for all $j \in M$. For sake of contradiction, suppose $p'_j > p_j$ for some $j \in M$. Let $S \subseteq M$ be the datasets with maximum multiplicative increase from $\mathbf{p}$ to $\mathbf{p}'$. By Claim 1, we have for every $j \in S$, if $(i, j) \in E_{\mathbf{p}'}$, then $(i, j) \in E_{\mathbf{p}}$, and $(i, j') \notin E_{\mathbf{p}}$ for $j' \notin S$.

Let $N'(S) = \{i \in N : x'_{i,j} > 0, j \in S\}$ and $N(S) = \{i \in N : x_{i,j} > 0, j \in S\}$ be the set of agents who buy datasets in S in $\mathbf{x}'$ and $\mathbf{x}$ respectively. The above claim implies that $N'(S) \subseteq N(S)$.

Observe that the market-clearing condition in $(\mathbf{p}', \mathbf{x}')$ implies that $\sum_{i \in N'(S)} b_i \geq \sum_{j \in S} p'_j$. The market-clearing condition in $(\mathbf{p}, \mathbf{x})$ and the fact that $(i, j') \notin E_{\mathbf{p}}$ for $i \in N(S)$ and $j' \notin S$ implies that $\sum_{j \in S} p_j \geq \sum_{i \in N(S)} b_i$. Putting the above observations with $p'_j > p_j$ for all $j \in S$, we obtain

$$\sum_{i \in N'(S)} b_i \geq \sum_{j \in S} p'_j > \sum_{j \in S} p_j \geq \sum_{i \in N(S)} b_i.$$

However, this cannot hold since $N'(S) \subseteq N(S)$. Thus, we must have $p'_j \leq p_j$ for all $j \in M$.

Let $\beta_i(\mathbf{p})$ and $\beta_i(\mathbf{p}')$ be the MBB-ratios of agent i in $(\mathbf{p}, \mathbf{x})$ and $(\mathbf{p}', \mathbf{x}')$ respectively. We clearly have $\beta_i(\mathbf{p}') \geq \beta_i(\mathbf{p})$ for all i . Using the definition of MBB-ratio and budget exhaustion, we have $\langle \tau_i, \mathbf{x}'_i \rangle = \beta_i(\mathbf{p}') \cdot \langle \mathbf{p}', \mathbf{x}'_i \rangle \geq \beta_i(\mathbf{p}) \cdot \langle \mathbf{p}, \mathbf{x}_i \rangle = \langle \tau_i, \mathbf{x}_i \rangle$. Using Eq. (2), we conclude that $u_i(\mathbf{x}'_i) \geq u_i(\mathbf{x}_i)$. $\square$

Multiplicity and Nonconvexity of CE for General Instances. We next demonstrate that uniqueness fails to hold in general. Even weaker structural properties, such as convexity— which are often desirable for computational reasons [23] – may not be guaranteed for the set of CE price vectors in general instances.

THEOREM 3.3. *There exists a data marketplace instance where there are multiple CE. Further, the set of CE is not convex.*

Rationality of CE Prices. We show that for non-degenerate instances with rational inputs, the CE prices are rational.

THEOREM 3.4. *For any non-degenerate data market instance with rational inputs, the CE prices are rational.*

Implications on computation of CE. Before presenting our algorithm, we provide a brief discussion on the implications of the results (Theorem 3.1, Theorem 3.3, and Theorem 3.4) on the computation of CE prices. At first glance, the uniqueness of CE in non-degenerate instances suggests the potential existence of an

underlying convex program, the global optimum of which corresponds to a CE (like the Eisenberg Gale convex program for Fisher markets [31]). However, any such program would need to leverage the non-degeneracy of the instance, as Theorem 3.3 establishes that a convex formulation is not possible in the degenerate case.

To the best of our knowledge, none of the well-known convex programs designed for CE in the classic rivalrous goods setting [23, 31, 50] explicitly incorporates non-degeneracy into their formulation. Even non-convex programs developed for related settings, such as the chores market [16], do not account for non-degeneracy in their design.

Secondly, in the rivalrous goods economy, allocation variables typically appear as dual variables in convex programs where prices are the primal variables (and vice versa). This dual structure ensures that, at equilibrium prices, the set of feasible allocations forms a convex cone. However, this property does not hold in our setting. In particular, the set of market-clearing allocations at a fixed price can still be non-convex. Consider the following example:

Example 3 (Non-convexity for market-clearing allocations at fixed price). *Consider a data market instance with four buyers, each with a budget of 1, and three sellers. We illustrate the values of $\tau_{i,j}$ in Fig. 1. If there is an edge from buyer i to seller j , then $\tau_{i,j} = 1$. Otherwise, $\tau_{i,j} = 0$. Consider two data allocations $\mathbf{x}^1$ and $\mathbf{x}^2$, represented by the bold edges in the picture (an edge is in bold if $x_{i,j} = 1$). The two allocations are both market-clearing. However, consider the average of the two allocations, $\mathbf{x}^3 = 0.5\mathbf{x}^1 + 0.5\mathbf{x}^2$. For seller 3, both $x^3_{3,3}$ and $x^3_{4,3}$ are equal to 0.5 while other $x^3_{1,3} = x^3_{2,3} = 0$, which violates the market-clearing condition.*

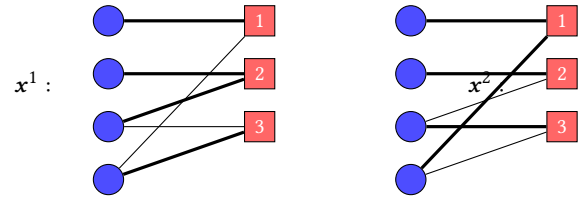


Figure 1: Two market-clearing allocations $\mathbf{x}^1$ and $\mathbf{x}^2$ for Example 3.

In summary, these observations highlight some interesting challenges that may arise while designing continuous optimization-based methods for computing CE prices, making this an intriguing direction for future work. In this paper, we instead develop a combinatorial algorithm that adjusts dataset prices based on supply and demand. At its core, the algorithm follows the *Walrasian* principle: “increase the prices of datasets where demand exceeds supply.” The main technical challenge lies in carefully identifying and maintaining certain invariants while updating the prices, which can guide the algorithm to the CE prices in polynomially many iterations.

4 Algorithm for Computing ϵ -CE

In this section, we present an FPTAS for computing an ϵ -approximate competitive equilibrium of a data marketplace, proving the following theorem.

THEOREM 4.1. *An ε -competitive equilibrium of a data marketplace can be computed in*

$$O\left(\frac{m}{\varepsilon} \cdot \log\left(\frac{2nb_{\max}\tau_{\max}}{b_{\min}\tau_{\min}}\right)\right) \text{ iterations.}$$

Recall from Lemma 2.3 that an ε -CE is a set of dataset prices $\mathbf{p}$ and an allocation $\mathbf{x}$ that together satisfy (i) budget exhaustion, (ii) maintain the MBB condition, and (iii) ensure market clearing.

Overview. At a high level, our algorithm follows a naturally intuitive auction-style *Walrasian* approach that iteratively increases the prices of the datasets whose demand exceeds their consumption. Our algorithm maintains the MBB condition and market-clearing condition as invariants and ensures that the spending of no buyer exceeds their budget. Our algorithm terminates as soon as all buyers approximately exhaust their budget. In every iteration of our algorithm, we perform two steps: (i) an *allocation update*, where we maximize the spending while keeping prices fixed, and (ii) a *price update*, where we increase the prices of those datasets that offer maximum bang-per-buck to the buyers who have the highest fraction of unspent budget.

The maintained invariants and termination condition imply that if the algorithm terminates, it does so with an approximate-CE. We show termination using a potential function argument that uses the fact that the algorithm monotonically increases prices of the datasets. This analysis relies critically on the algorithm operating over instances with certain structural properties. We show that such instances can be obtained from the original instance via mild perturbations. Furthermore, we prove that any ε -CE obtained in a perturbed instance corresponds to an $O(\varepsilon)$ -CE in the original instance. Therefore, it suffices to only focus on perturbed instances, as described below.

Definition 4.2 (ε -Perturbed Instance). Given a data market instance $\mathcal{I}$ with parameters $\{\tau_{i,j}\}_{i \in N, j \in M}$ and a parameter $\varepsilon \in (0, \frac{1}{10}]$, we construct an perturbed instance $\mathcal{I}'$ with parameters $\{\tau'_{i,j}\}_{i \in N, j \in M}$ as follows. We let $\tau'_{i,j} = 0$ whenever $\tau_{i,j} = 0$. Otherwise, we set $\tau'_{i,j}$ to be the smallest integral power of $(1 - \varepsilon)^{-1}$ that is at least $\tau_{i,j}$. That is, $\tau_{i,j} = (1 - \varepsilon)^{-\alpha_{i,j}}$, where $\alpha_{i,j} = \lceil \log_{(1-\varepsilon)^{-1}} \tau_{i,j} \rceil$.

We now argue that an ε -CE in the perturbed instance $\mathcal{I}'$ is a 3ε -CE in the original instance $\mathcal{I}$.

LEMMA 4.3. *Let $(\mathbf{p}, \mathbf{x})$ be an ε -CE in the perturbed instance $\mathcal{I}'$. Then $(\mathbf{p}, \mathbf{x})$ is a 3ε -CE in the original instance $\mathcal{I}$.*

PROOF. We show that $(\mathbf{p}, \mathbf{x})$ is a 3ε -CE in $\mathcal{I}$ using the characterization of approximate CE from Lemma 2.3.

- (Approximate budget exhaustion) For each buyer $i \in N$, we have $(1 - \varepsilon) \cdot b_i \leq \langle \mathbf{p}, \mathbf{x}_i \rangle \leq b_i$ since $(\mathbf{p}, \mathbf{x})$ is an ε -CE in $\mathcal{I}$.
- (Approximate MBB) Let $G'_p(\varepsilon) = (V, E'_p(\varepsilon))$ be the ε -MBB-graph in instance $\mathcal{I}'$ at prices $\mathbf{p}$. Consider $(i, j) \in N \times M$ such that $x_{i,j} > 0$. Since $(\mathbf{p}, \mathbf{x})$ is an ε -CE, we have $(i, j) \in E'_p(\varepsilon)$, i.e., $\frac{\tau'_{i,j}}{p_j} \geq (1 - \varepsilon) \cdot \max_k \frac{\tau'_{i,k}}{p_k}$. Since $\tau_{i,j} \leq \tau'_{i,j} \leq (1 - \varepsilon)^{-1} \tau'_{i,j}$,

$$\frac{\tau_{i,j}}{p_j} \geq (1 - \varepsilon) \cdot \frac{\tau'_{i,j}}{p_j} \geq (1 - \varepsilon)^2 \cdot \max_k \frac{\tau'_{i,k}}{p_k} \geq (1 - 2\varepsilon) \cdot \max_k \frac{\tau_{i,k}}{p_k}.$$

This shows that $x_{i,j} > 0$ only if $(i, j) \in E_p(2\varepsilon)$, where $G_p(2\varepsilon)$ is the 2ε -MBB-graph at prices $\mathbf{p}$ in instance $\mathcal{I}$.

Algorithm 1: FPTAS for computing ε -competitive equilibrium

```

1 ▶ Initial pricing and allocation
2 Initialize  $x_{i,j} = 0$  for all  $i \in N, j \in M$ 
3 for each  $j \in M$  do
4   | Set  $p_j \leftarrow \tau_{i,j}$  for  $i \in \arg \max_{k \in N} \tau_{k,j}$  and  $x_{i,j} \leftarrow 1$ ;
5 Set  $\delta = \max_{i \in N} \langle \mathbf{p}, \mathbf{x}_i \rangle / b_i$ ;
6 if  $\delta > 1$  then
7   | for each  $j \in M$  do
8     |  $p_j \leftarrow (1 - \varepsilon)^{-\lambda_j}$  where  $\lambda_j = \lceil \log_{(1-\varepsilon)^{-1}} (p_j / \delta) \rceil$ ;
9 while there exists a buyer  $i$  with surplus  $\Delta_i(\mathbf{p}, \mathbf{x}) \geq \varepsilon \cdot b_i$  do
10  | ▶ Allocation update
11  | while there exists  $x_{i,j} < 1$  with  $\Delta_i(\mathbf{p}, \mathbf{x}) > 0$  and  $(i, j)$ 
12    | being an MBB edge do
13    | Increase  $x_{i,j}$  until  $x_{i,j} = 1$  or  $\Delta_i(\mathbf{p}, \mathbf{x})$  becomes zero;
14  | if no buyer has surplus  $\Delta_i(\mathbf{p}, \mathbf{x}) \geq \varepsilon \cdot b_i$  then
15    | break;
16  | ▶ Price update
17  | Let  $B$  be the set of buyers whose surplus  $\Delta_i(\mathbf{p}, \mathbf{x})$  has
18    | the largest proportion to their budget, i.e.,  $\Delta_i(\mathbf{p}, \mathbf{x}) / b_i$ 
19    | is maximized;
20  | Let  $N_B$  be the set of sellers that are adjacent to  $B$  in the
21    | MBB graph;
22  | Increase the price for every dataset  $\mathcal{D}_j$  with  $j \in N_B$  by
23    | the same factor  $(1 - \varepsilon)^{-1}$ ;
24  | for every buyer  $i \in [n] \setminus B$  do
25    | Reduce  $x_{i,j}$  to zero if some dataset  $j \in N_B$  is no
26    | longer an MBB dataset for buyer  $i$  at new price  $\mathbf{p}$ ;
27 return  $\mathbf{p}$  and  $\mathbf{x}$ ;

```

- (Market-clearing) Since $(\mathbf{p}, \mathbf{x})$ satisfies the market clearing condition in $\mathcal{I}'$, it does so in $\mathcal{I}$ as well.

We invoke Lemma 2.3 and conclude that $(\mathbf{p}, \mathbf{x})$ is a 3ε -CE in $\mathcal{I}$. $\square$

4.1 Computing ε -CE in Perturbed Instances

We now describe our algorithm for finding an ε -CE for a perturbed instance. The pseudo-code is presented in Algorithm 1. Our algorithm begins with an initialization phase that constructs an initial set of prices $\mathbf{p}^0$ and an initial allocation $\mathbf{x}^0$. Each subsequent iteration of the algorithm has (i) an allocation update step that modifies the current allocation $\mathbf{x}$ while keeping the current prices $\mathbf{p}$ the same, and (ii) a price update step that modifies $\mathbf{p}$ and $\mathbf{x}$. Throughout, our algorithm maintains the following properties as invariants:

Invariant 1. *Let $\mathbf{p}$ and $\mathbf{x}$ be the price vector and allocation maintained by algorithm 1 in some iteration. Then:*

- (i) *The price p_j of each dataset $\mathcal{D}_j$ is an integral power of $(1 - \varepsilon)^{-1}$.*
- (ii) *$(\mathbf{p}, \mathbf{x})$ satisfy the MBB condition, i.e., $x_{i,j} > 0$ implies $(i, j) \in E_p$.*
- (iii) *Market-clearing holds for every dataset, i.e., $\max_i x_{i,j} = 1 \forall j \in M$.*
- (iv) *No buyer overspends, i.e., $\langle \mathbf{p}, \mathbf{x}_i \rangle \leq b_i$ for all $i \in N$.*

We now describe the phases in detail and their properties.

4.1.1 Initializing Prices. We first find an initial price vector $\mathbf{p}^0$ and an allocation $\mathbf{x}^0$ that together satisfy Invariant 1. We construct a

preliminary price vector $\mathbf{p}$ by finding for each dataset $j \in M$, a buyer $i \in \arg \max_{k \in N} \tau_{k,j}$, and setting $p_j = \tau_{i,j}$ and $x_{i,j} = 1$.

First observe that for every $j \in M$ there exists an $i \in N$ s.t. $\tau_{i,j} > 0$, showing that each $p_j > 0$. Next, note that the pair $(\mathbf{p}, \mathbf{x})$ satisfies condition (i) of Invariant 1 since all $\tau_{i,j}$'s are integral powers of $(1 - \varepsilon)^{-1}$. Now consider any (i, j) such that $x_{i,j} > 0$. We have $\tau_{i,j}/p_j = 1$ and $\tau_{i,\ell}/p_\ell \leq \frac{\tau_{i,\ell}}{\max_{k \in N} \tau_{k,\ell}} \leq 1$ for any $\ell \in M$. Thus $x_{i,j} > 0$ implies $(i, j) \in E_{\mathbf{p}}$ for every (i, j) , showing that condition (ii) of Invariant 1 is satisfied. Moreover, the market-clearing condition (iii) is also satisfied for every dataset by construction.

Let $\delta = \max_{i \in N} \langle \mathbf{p}, \mathbf{x}_i \rangle / b_i$. If $\delta \leq 1$, condition (iv) of Invariant 1 is also satisfied, and we simply set $\mathbf{p}^0 \leftarrow \mathbf{p}$ as the initial price vector and $\mathbf{x}$ as the initial allocation. On the other hand, if $\delta > 1$, some buyers overspend. We uniformly scale down all prices to attain condition (iv). To simultaneously maintain condition (i), we set $p_j^0 = (1 - \varepsilon)^{-\lambda_j}$, where $\lambda_j = \lfloor \log_{(1-\varepsilon)^{-1}}(p_j/\delta) \rfloor$. Note that for every $p_j^0 > 0$ for every $j \in M$. Further, conditions (i), (ii), and (iii) are still satisfied by $(\mathbf{p}^0, \mathbf{x})$. To see why condition (iv) holds, note that $p_j^0 \leq p_j/\delta$ for every $j \in M$ by construction. Then by definition of δ , we have $\langle \mathbf{p}^0, \mathbf{x}_i \rangle \leq \langle \mathbf{p}, \mathbf{x}_i \rangle / \delta \leq b_i$ for every buyer i .

This concludes the initialization phase, resulting in an initial price vector $\mathbf{p}^0 > 0$ and an allocation $\mathbf{x}$ that together satisfy Invariant 1. We record the above observations and an additional important property of the initial prices in the following lemma.

LEMMA 4.4. *The initial prices $\mathbf{p}^0$ and allocation $\mathbf{x}$ satisfy Invariant 1 and can be computed in polynomial time. Moreover, $p_j^0 \geq \frac{(1-\varepsilon)\tau_{\min}b_{\min}}{m\tau_{\max}}$ for every $j \in M$.*

PROOF. The fact that $(\mathbf{p}^0, \mathbf{x})$ satisfies Invariant 1 and can be computed in polynomial time is clear from the preceding discussion. For the lower bound on prices, note that $p_j^0 \geq (1 - \varepsilon)p_j/\delta$, where $p_j = \tau_{i,j}$ for some $i \in N$. Note that $p_j \geq \tau_{\min} > 0$ for every $j \in M$. Next, observe that $\delta = \max_{k \in N} \frac{\langle \mathbf{p}, \mathbf{x}_k \rangle}{b_k} \leq \frac{m\tau_{\max}}{b_{\min}}$. Thus, we obtain that $p_j^0 \geq \frac{(1-\varepsilon)\tau_{\min}b_{\min}}{m\tau_{\max}}$ for every $j \in M$. $\square$

4.1.2 Allocation and Price Updates. Having computed the initial prices and allocation, the algorithm iteratively updates the prices vector $\mathbf{p}$ and allocation $\mathbf{x}$ until an $O(\varepsilon)$ -CE is reached. More precisely, the algorithm terminates as soon as the *surplus* $\Delta_i(\mathbf{p}, \mathbf{x}) := b_i - \langle \mathbf{p}, \mathbf{x}_i \rangle$ is at most $\varepsilon \cdot b_i$ for every buyer i . While there exists buyers i with $\Delta_i(\mathbf{p}, \mathbf{x}) > \varepsilon \cdot b_i$, we perform two types of updates in each iteration to clear the remaining budgets: *allocation update* and *price update*.

Allocation update: The goal of this step is to maximize the spending of buyers at the current prices while maintaining feasibility. Thus, if there exists a buyer i with positive surplus who can increase spending on an MBB-dataset j , we do so until either j is fully consumed by i or the budget of i exhausts. That is, for a buyer i with $\Delta_i(\mathbf{p}, \mathbf{x}) > 0$ such that $(i, j) \in E_{\mathbf{p}}$ and $x_{i,j} < 1$ for some $j \in M$, we increase $x_{i,j}$ until either $x_{i,j} = 1$ or the $\langle \mathbf{p}, \mathbf{x}_i \rangle = b_i$.

Therefore, after the allocation update, for every buyer i , either $\Delta_i(\mathbf{p}, \mathbf{x}) = 0$ or $x_{i,j} = 1$ for every MBB-dataset j of buyer i .

Price update: The goal of this step is to increase prices of the datasets whose demand exceeds their consumption, and subsequently also adjust the allocation. We identify the set of buyers B

whose surplus $\Delta_i(\mathbf{p}, \mathbf{x})$ has the largest proportion to their budget, i.e., $\Delta_i(\mathbf{p}, \mathbf{x})/b_i$ is maximum. Let $N_B = \{j \in M : (i, j) \in E_{\mathbf{p}} \text{ for } i \in B\}$ be the set of sellers that are adjacent to B in the current MBB graph $E_{\mathbf{p}}$. Then, we increase the price for every dataset $\mathcal{D}_j$ with $j \in N_B$ by the same factor of $(1 - \varepsilon)^{-1}$. Due to this update, a dataset $j \in N_B$ may no longer be an MBB-dataset for a buyer $i \in N \setminus B$ at the new prices. If this happens, we update the set the corresponding allocation as $x_{i,j} = 0$.

LEMMA 4.5. *The allocation and price updates maintain Invariant 1.*

PROOF. We prove the four properties of Invariant 1 below.

- (i) Since the initial prices are integral powers of $(1 - \varepsilon)^{-1}$ and the price update step performs multiplicative increments in powers of $(1 - \varepsilon)^{-1}$, the prices are always integral powers of $(1 - \varepsilon)^{-1}$. Note that the allocation update does not change the prices.
- (ii) The allocation update step does not decrease $x_{i,j}$ for any i, j , and hence maintains the MBB invariant. We then consider the price update step which updates the price vector $\mathbf{p}$ to $\mathbf{p}'$. For a buyer $i \notin B$, the MBB condition is maintained at $\mathbf{p}'$ since we ensure $x_{i,j} = 0$ for $j \notin E_{\mathbf{p}'}$ (line 20). Consider a buyer $i \in B$ and define $\gamma_i := \max_{\ell \in M} \tau_{i,\ell}/p_\ell$. By definition of N_B , we have $\tau_{i,j}/p_j = \gamma_i$ for all $j \in N_B$ such that $x_{i,j} > 0$, and $\tau_{i,k}/p_k < \gamma_i$ for all $k \notin N_B$. Since the parameters $\tau_{i,j}$ and the prices are integral powers of $(1 - \varepsilon)^{-1}$, we in fact have $\tau_{i,k}/p_k \leq (1 - \varepsilon)\gamma_i$ for all $k \notin N_B$. After the price update, we have $\tau_{i,j}/p'_j = (1 - \varepsilon)\gamma_i$ for every $j \in N_B$ such that $x_{i,j} > 0$, whereas $\tau_{i,k}/p'_k = \tau_{i,k}/p_k \leq (1 - \varepsilon)\gamma_i$. Thus, we have $x_{i,j} > 0$ only if $(i, j) \in E_{\mathbf{p}'}$, showing that the MBB condition is maintained for all buyers at the updated prices $\mathbf{p}'$.
- (iii) The allocation update step does not decrease $x_{i,j}$ for any i, j , and hence maintains the market-clearing invariant. The price update step can only reduce $x_{i,j}$ for a buyer $i \in N \setminus B$ and dataset $j \in N_B$ (line 20). However, the preceding allocation update step ensures that there is some buyer $i' \in B$ such that $x_{i',j} = 1$. Thus the market-clearing condition continues to hold for j as well.
- (iv) The allocation update step only continues until $\Delta_i(\mathbf{p}, \mathbf{x}) = 0$ and does not cause any buyer to overspend. We then consider the price update step. Note that the spending of a buyer $i \notin B$ cannot increase. For a buyer $i \in B$, we have $\Delta_i(\mathbf{p}, \mathbf{x}) > \varepsilon \cdot b_i$, otherwise the algorithm would have terminated. Thus, $\langle \mathbf{p}, \mathbf{x}_i \rangle < (1 - \varepsilon) \cdot b_i$. Therefore, the spending of i at the updated prices $\mathbf{p}'$ satisfies $\langle \mathbf{p}', \mathbf{x}_i \rangle = (1 - \varepsilon)^{-1} \cdot \langle \mathbf{p}, \mathbf{x}_i \rangle < b_i$. $\square$

Having established Invariant 1, it becomes clear that if Algorithm 1 terminates, it returns an ε -CE. We prove the termination and analyze the time-complexity of our algorithm in Theorem D.1 using a potential function argument.

We remark that the existence of *exact* competitive equilibrium follows from the existence of ε -CE together with compactness arguments (see Remark 2).

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References

- [1] ACEMOGLU, D., MAKHDOUNI, A., MALEKIAN, A., AND OZDAGLAR, A. When big data enables behavioral manipulation. *American Economic Review: Insights* 7, 1 (2025), 19–38.
- [2] ADMATI, A. R., AND PFLEIDERER, P. A monopolistic market for information. *Journal of Economic Theory* 39, 2 (1986), 400–438.
- [3] ADMATI, A. R., AND PFLEIDERER, P. Direct and indirect sale of information. *Econometrica: Journal of the Econometric Society* (1990), 901–928.
- [4] AGARWAL, A., DAHLEH, M. A., HOREL, T., AND RUI, M. Towards data auctions with externalities. *Games Econ. Behav.* 148 (2024), 323–356.
- [5] AGARWAL, A., DAHLEH, M. A., AND SARKAR, T. A marketplace for data: An algorithmic solution. In *EC* (2019), ACM, pp. 701–726.
- [6] AKRAMI, H., CHAUDHURY, B. R., GARG, J., AND MURHEKAR, A. On the theoretical foundations of data exchange economies. In *Proceedings of the 26th ACM Conference on Economics and Computation* (New York, NY, USA, 2025), EC '25, Association for Computing Machinery, p. 444.
- [7] ARROW, K. J., AND DEBREU, G. Existence of an equilibrium for a competitive economy. *Econometrica: Journal of the Econometric Society* (1954), 265–290.
- [8] BABAOFF, M., KLEINBERG, R., AND LEME, R. P. Optimal mechanisms for selling information. In *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)* (2012), p. 92–109.
- [9] BALEY, I., AND VELDkamp, L. L. The data economy: Tools and applications. In *The Data Economy*. Princeton University Press, 2025.
- [10] BEI, X., GARG, J., AND HOEFER, M. Ascending-price algorithms for unknown markets. *ACM Trans. Algorithms* (2019).
- [11] BERGEMANN, D., AND BONATTI, A. Data, competition, and digital platforms. *American Economic Review* 114, 8 (2024), 2553–2595.
- [12] BERGEMANN, D., BONATTI, A., AND GAN, T. The economics of social data. *The RAND Journal of Economics* 53, 2 (2022), 263–296.
- [13] BERGEMANN, D., BONATTI, A., AND SMOLIN, A. The design and price of information. *American economic review* 108, 1 (2018), 1–48.
- [14] BHASKARA, A., GOLLAPUDI, S., IM, S., KOLLIAS, K., MUNAGALA, K., AND SANKAR, G. S. Data exchange markets via utility balancing. In *WWW* (2024), ACM, pp. 57–65.
- [15] CAI, Y., AND VELEGKAS, G. How to sell information optimally: An algorithmic study. In *12th Innovations in Theoretical Computer Science Conference, ITCS* (2021).
- [16] CHAUDHURY, B. R., KROER, C., MEHTA, R., AND NAN, T. Competitive equilibrium for chores: from dual Eisenberg-Gale to a fast, greedy, LP-based algorithm. *arXiv preprint arXiv:2402.10439* (2024).
- [17] CHEN, X., DAI, D., DU, Y., AND TENG, S. Settling the complexity of Arrow-Debreu equilibria in markets with additively separable utilities. In *Proc. 50th Symposium Foundations of Computer Science (FOCS)* (2009), pp. 273–282.
- [18] CHEN, X., HUANG, L.-S., AND TENG, S.-H. Market equilibria with hybrid linear-Leontief utilities. In *WINE* (2006), pp. 274–285.
- [19] CHEN, X., PAPARAS, D., AND YANNAKAKIS, M. The complexity of non-monotone markets. *Journal of the ACM (JACM)* 64, 3 (2017), 1–56.
- [20] CHEN, X., AND TENG, S. Spending is not easier than trading: On the computational equivalence of Fisher and Arrow-Debreu equilibria. In *Proc. 20th International Symposium Algorithms and Computation (ISAAC)* (2009), pp. 647–656.
- [21] CODENOTTI, B., PEMMARAJU, S., AND VARADARAJAN, K. On the polynomial time computation of equilibria for certain exchange economies. In *Proceedings of the 16th ACM-SIAM Symposium on Discrete Algorithms (SODA)* (2005), pp. 72–81.
- [22] CODENOTTI, B., SABERI, A., VARADARAJAN, K., AND YE, Y. Leontief economies encode nonzero sum two-player games. In *SODA* (2006), vol. 6, pp. 659–667.
- [23] COLE, R., DEVANUR, N., GKATZELIS, V., JAIN, K., MAI, T., VAZIRANI, V., AND YAZDANBOD, S. Convex program duality, Fisher markets, and Nash social welfare. In *Proc. 18th Conf. Economics and Computation (EC)* (2017).
- [24] COLE, R., AND FLEISCHER, L. Fast-converging tatonnement algorithms for one-time and ongoing market problems. In *Proceedings of the Fortieth Annual ACM Symposium on Theory of Computing (STOC)* (2008), p. 315–324.
- [25] CUMMINGS, R., ELZAYN, H., POUNTOURAKIS, E., GKATZELIS, V., AND ZIANI, J. Optimal data acquisition with privacy-aware agents. In *2023 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)* (2023), IEEE, pp. 210–224.
- [26] CUMMINGS, R., LIGETT, K., ROTH, A., WU, Z. S., AND ZIANI, J. Accuracy for sale: Aggregating data with a variance constraint. In *Proceedings of the 2015 conference on innovations in theoretical computer science (ITCS)* (2015), pp. 317–324.
- [27] DELIGKAS, A., FEARNEY, J., HOLLENDER, A., AND MELISSOURGOS, T. Constant inapproximability for Fisher markets. In *Proc. 25th Conf. Economics and Computation (EC)* (2024).
- [28] DEVANUR, N., PAPADIMITRIOU, C., SABERI, A., AND VAZIRANI, V. Market equilibrium via a primal–dual algorithm for a convex program. *J. ACM* 55, 5 (2008).
- [29] DUAN, R., GARG, J., AND MEHLHORN, K. An improved combinatorial polynomial algorithm for the linear arrow-debreu market. In *SODA* (2016), SIAM, pp. 90–106.
- [30] EISENBERG, E. Aggregation of utility functions. *Management Sci.* 7, 4 (1961), 337–350.
- [31] EISENBERG, E., AND GALE, D. Consensus of subjective probabilities: The Pari-Mutuel method. *Ann. Math. Stat.* 30, 1 (1959), 165–168.
- [32] FAJGELBAUM, P. D., SCHAAL, E., AND TASCHEREAU-DUMOUCHEL, M. Uncertainty traps. *The Quarterly Journal of Economics* 132, 4 (2017), 1641–1692.
- [33] FALLAH, A., JORDAN, M. I., MAKHDOUNI, A., AND MALEKIAN, A. On three-layer data markets. *arXiv preprint arXiv:2402.09697* (2024).
- [34] FALLAH, A., MAKHDOUNI, A., MALEKIAN, A., AND OZDAGLAR, A. Bridging central and local differential privacy in data acquisition mechanisms. *Advances in Neural Information Processing Systems* 35 (2022), 21628–21639.
- [35] FALLAH, A., MAKHDOUNI, A., MALEKIAN, A., AND OZDAGLAR, A. Optimal and differentially private data acquisition: Central and local mechanisms. *Operations Research* 72, 3 (2024), 1105–1123.
- [36] FARBOODI, M., SINGAL, D., VELDkamp, L., AND VENKATESWARAN, V. Valuing financial data. *The Review of Financial Studies* 38, 3 (2025), 938–980.
- [37] FARBOODI, M., AND VELDkamp, L. Data and markets. *Annual Review of Economics* 15, 1 (2023), 23–40.
- [38] FLECKENSTEIN, M., OBAIDI, A., AND TRYFONA, N. A review of data valuation approaches and building and scoring a data valuation model. *Harvard Data Science Review* 5, 1 (2023).
- [39] GARG, J., MEHTA, R., VAZIRANI, V. V., AND YAZDANBOD, S. Settling the complexity of Leontief and PLC exchange markets under exact and approximate equilibria. In *Symp. Theory of Computing (STOC)* (2017), pp. 890–901.
- [40] HOSSAIN, S., AND CHEN, Y. Equilibrium of data markets with externality. In *ICML* (2024), OpenReview.net.
- [41] JAIN, K., AND VAZIRANI, V. Equilibrium pricing of semantically substitutable digital goods. *arXiv preprint arXiv:1007.4586* (2010).
- [42] MAS-COLELL, A., WHINSTON, M. D., GREEN, J. R., ET AL. *Microeconomic theory*, vol. 1. Oxford university press New York, 1995.
- [43] MEHTA, S., DAWANDE, M., JANAKIRAMAN, G., AND MOOKERJEE, V. How to sell a data set? pricing policies for data monetization. *Information Systems Research* 32, 4 (2021), 1281–1297.
- [44] MURHEKAR, A., YUAN, Z., CHAUDHURY, B. R., LI, B., AND MEHTA, R. Incentives in federated learning: Equilibria, dynamics, and mechanisms for welfare maximization. In *NeurIPS* (2023).
- [45] ORDONEZ, G. The asymmetric effects of financial frictions. *Journal of Political Economy* 121, 5 (2013), 844–895.
- [46] ORLIN, J. Improved algorithms for computing Fisher’s market clearing prices. In *Proc. 42nd Symp. Theory of Computing (STOC)* (2010), pp. 291–300.
- [47] PEI, J. A survey on data pricing: from economics to data science. *IEEE Transactions on Knowledge and Data Engineering* 34, 10 (2020), 4586–4608.
- [48] RESEARCH, A. Big data market size: Global industry, share, analysis, trends and forecast 2022 - 2030.
- [49] RUBINSTEIN, A. Inapproximability of nash equilibrium. *SIAM J. Comput.* 47, 3 (2018), 917–959.
- [50] SHMYREV, V. An algorithm for finding equilibrium in the linear exchange model with fixed budgets. *J. Appl. Indust. Math.* 3, 4 (2009), 505–518.
- [51] SONG, J., KULKARNI, P., SHAHKAR, P., AND CHAUDHURY, B. On the existence and complexity of core-stable data exchanges. In *Proceedings of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS)* (2025).
- [52] VÉGH, L. A. A strongly polynomial algorithm for a class of minimum-cost flow problems with separable convex objectives. *SIAM J. Comput.* 45, 5 (2016), 1729–1761.
- [53] VELDkamp, L. Valuing data as an asset. *Review of Finance* 27, 5 (2023), 1545–1562.
- [54] VELDkamp, L. L. Slow boom, sudden crash. *Journal of Economic theory* 124, 2 (2005), 230–257.
- [55] ZHANG, L. Proportional response dynamics in the Fisher market. *Theoretical Computer Science* 412, 24 (2011), 2691–2698.

A Discussion

In this paper, we propose a model for the competitive pricing of data, building on classic equilibrium theory [7] and the economic value of data [9]. Our work represents a first step towards understanding competitive equilibria in data markets, and opens up several directions for future research, including extending the model to capture richer utility functions, as well as technical questions within the current framework.

Algorithms and dynamics. A natural next step is to investigate the computation of *exact* competitive equilibria (CE) in our model. Classical algorithms for computing CE, such as those for Fisher and Arrow–Debreu markets [7, 28, 46, 52], critically rely on iterative reductions (of a function monotonic in) excess budget to guarantee convergence.

Another fundamental question is the existence of *natural dynamics*, such as *Tatonnement* [24] or *Proportional Response* [55], that converge to a CE. While such convergence is typically shown by establishing an equivalence between the dynamics and first-order methods applied to convex programs like the Eisenberg–Gale formulation [31], our results indicate that an analogous convex program is unlikely to exist for our setting. Nonetheless, our Walrasian-style price adjustment algorithm, which mirrors tatonnement, suggests that these dynamics may still converge, although we suspect that proving this will likely require non-standard techniques.

Modeling generalizations and extensions. A natural generalization of our model is to consider the richer class of interactions between the data buyers and sellers arising from *non-linear correlations* among the data records of different sellers. This phenomenon is common in domains such as medical diagnostics, where results from multiple tests can uncover far more about a patient’s underlying condition than any single test in isolation. We suspect that a CE still exists in such settings, given that the utility functions resemble the well-studied class of *linear Leontief utilities* [18], which are known to admit CE in conventional economies. However, computing a CE may be challenging without a convex programming formulation.

Another extension involves the prediction task of the buyer involves *multiple states* with varying utilities and risks. For instance, in financial prediction tasks, buyers may evaluate trade-offs among different data sellers in terms of expected gains from data against the uncertainty in their predictions. Exploring the structure and computation of CE under such multi-dimensional utilities represents a natural and important continuation of this work.

B Omitted Proofs from Section 2

LEMMA 2.3. *Let $\varepsilon_1, \varepsilon_2 > 0$ be parameters. Suppose $(\mathbf{p}, \mathbf{x})$ satisfies the following conditions:*

- (i) *(Approximate budget exhaustion) For each buyer $i \in N$, $(1 - \varepsilon_1) \cdot b_i \leq \langle \mathbf{p}, \mathbf{x}_i \rangle \leq b_i$.*
- (ii) *(Approximately maximum bang-per-buck) For each buyer $i \in N$ and seller $j \in M$, $x_{i,j} > 0$ only if $(i, j) \in E_{\mathbf{p}}(\varepsilon_2)$.*
- (iii) *(Market-clearing) For each seller $j \in M$, $\max_{i \in N} x_{i,j} = s_j$.*

Then $(\mathbf{p}, \mathbf{x})$ is an $(\varepsilon_1 + \varepsilon_2)$ -CE.

PROOF. We first show that $(\mathbf{p}, \mathbf{x})$ satisfies the approximately-optimal bundle condition from Definition 2.1. Consider a buyer $i \in N$, and let $\mathbf{y}$ be any allocation such that $\langle \mathbf{p}, \mathbf{y}_i \rangle \leq b_i$. We observe that:

$$\begin{aligned}
 \sum_{j=1}^m \tau_{i,j} \cdot x_{i,j} &= \sum_{j=1}^m \frac{\tau_{i,j}}{p_j} \cdot (p_j \cdot x_{i,j}) \\
 &\geq (1 - \varepsilon_2) \max_{j \in M} \frac{\tau_{i,j}}{p_j} \cdot \langle \mathbf{p}, \mathbf{x}_i \rangle \quad (\text{using condition (ii)}) \\
 &\geq (1 - \varepsilon_2) \max_{j \in M} \frac{\tau_{i,j}}{p_j} \cdot (1 - \varepsilon_1) b_i \quad (\text{using condition (i)}) \\
 &\geq (1 - \varepsilon_1 - \varepsilon_2) \max_{j \in M} \frac{\tau_{i,j}}{p_j} \langle \mathbf{p}, \mathbf{y}_i \rangle \quad (\text{since } \langle \mathbf{p}, \mathbf{y}_i \rangle \leq b_i) \\
 &\geq (1 - \varepsilon_1 - \varepsilon_2) \sum_{j \in M} \tau_{i,j} \cdot y_{i,j}.
 \end{aligned}$$

With τ_i denoting the vector $(\tau_{i1}, \tau_{i2}, \dots, \tau_{im})$, the above inequalities show that $\langle \tau_i, \mathbf{x}_i \rangle \geq (1 - \varepsilon_1 - \varepsilon_2) \cdot \langle \tau_i, \mathbf{y}_i \rangle$. We then observe

that:

$$\frac{u_i(\mathbf{x})}{u_i(\mathbf{y})} = \frac{\sum_{j=1}^m \tau_{i,j} \cdot x_{i,j}}{\sum_{j=1}^m \tau_{i,j} \cdot y_{i,j}} \geq (1 - \varepsilon_1 - \varepsilon_2).$$

Therefore, every agent gets a bundle in $\mathbf{x}$ that she values at least $(1 - \varepsilon_1 - \varepsilon_2)$ times the optimal feasible utility at prices $\mathbf{p}$. Further, since condition (iii) is the market-clearing condition, we conclude that $(\mathbf{p}, \mathbf{x})$ is an $(\varepsilon_1 + \varepsilon_2)$ -CE. $\square$

Normalization of supply. We now formally prove that normalizing supply and setting $s_j = 1$ for all j is without loss of generality. Let $\mathcal{I}$ be a data market instance with parameters $\{\tau_i\}_{i \in N}$, $\{\tau_{i,j}\}_{i \in N, j \in M}$, and $\{s_j\}_{j \in M}$. We construct a normalized instance $\mathcal{I}'$ with parameters $\{\tau'_i\}_{i \in N}$, $\{\tau'_{i,j}\}_{i \in N, j \in M}$, and $\{s'_j\}_{j \in M}$ by setting $\tau'_i = \tau_i$ and $\tau'_{i,j} = \tau_{i,j} \cdot s_j$ and $s'_j = 1$, for every $i \in N$ and $j \in M$.

LEMMA B.1. *For any $\varepsilon > 0$, if $(\mathbf{p}', \mathbf{x}')$ is an ε -CE of the normalized data market instance $\mathcal{I}'$ after supply normalization, then $(\mathbf{p}, \mathbf{x})$ with $p_j = p'_j / s_j$ and $x_{i,j} = s_j \cdot x'_{i,j}$ for every $i \in N, j \in M$ is an ε -CE of the original data market instance $\mathcal{I}$.*

PROOF. Consider any allocation $\mathbf{y}$ of $\mathcal{I}$, and let $\mathbf{y}'$ be an allocation of $\mathcal{I}'$ given by $y'_{i,j} = y_{i,j} / s_j$. Observe that by construction of $\mathcal{I}'$, we have $\tau'_i = \tau_i$ and $\tau'_{i,j} \cdot y'_{i,j} = \tau_{i,j} \cdot y_{i,j}$. Thus $u'_i(\mathbf{y}') = u_i(\mathbf{y})$, i.e., each buyer gets identical utilities from allocations $\mathbf{y}'$ and $\mathbf{y}$ of markets $\mathcal{I}'$ and $\mathcal{I}$ respectively. Further, by definition of $\mathbf{p}'$, we have $p_j \cdot y_{i,j} = p'_j \cdot y'_{i,j}$ implying that $\mathbf{y}_i$ is feasible at prices $\mathbf{p}$ in $\mathcal{I}$ iff $\mathbf{y}'_i$ is feasible at prices $\mathbf{p}'$ in $\mathcal{I}'$.

The above observations imply that for each agent $i \in N$, $\mathbf{x}_i$ is an ε -optimal bundle at prices $\mathbf{p}$ iff $\mathbf{x}'_i$ is an ε -optimal bundle at prices $\mathbf{p}'$. Moreover, $\mathbf{y}$ clears the market for $\mathcal{I}$ iff $\mathbf{y}'$ clears the market for $\mathcal{I}'$, since $\max_{i \in N} y'_{i,j} = 1$ iff $\max_{i \in N} y_{i,j} = s_j$ for each $j \in M$.

In conclusion, we obtain that $(\mathbf{p}, \mathbf{x})$ is an ε -CE of $\mathcal{I}$ if $(\mathbf{p}', \mathbf{x}')$ is an ε -CE of $\mathcal{I}'$, as claimed. $\square$

C Omitted Proofs from Section 3

C.1 Rationality of CE prices

THEOREM 3.4. *For any non-degenerate data market instance with rational inputs, the CE prices are rational.*

PROOF. Let $(\mathbf{p}, \mathbf{x})$ be a CE of a non-degenerate instance with rational inputs. Recall that $G_{\mathbf{p}}$ is acyclic since the instance is non-degenerate. Consider any connected component T of $G_{\mathbf{p}}$, and select an arbitrary dataset j_0 in T . We first show that we can express the prices of all datasets in T as rational multiples of p_{j_0} . By writing the MBB condition along the edges of T , we obtain that $p_j = r_j \cdot p_{j_0}$ for any j in T , where r_j is a ratio of products of the input parameters $\tau_{k,\ell}$. Since the input is rational, we conclude that $r_j \in \mathbb{Q}^+$ for all j in T .

Next, we show that there must be some agent i in T who does not receive any partial datasets, i.e., there is no j s.t. $x_{i,j} \in (0, 1)$. To see why, pick any arbitrary agent i_1 in T . If i_1 does not have a partial dataset, we are done. Otherwise, $x_{i_1, j_1} \in (0, 1)$ for some j_1 in T . By the market-clearing condition, there must be some agent $i_2 \neq i_1$ s.t. $x_{i_2, j_1} = 1$. Again, if i_2 has a partial dataset j_3 , then there must be an agent i_3 s.t. $x_{i_3, j_3} = 1$. Continuing this argument either results in an agent i with no partial datasets, or a cycle in $G_{\mathbf{p}}$. Since the latter

is not possible due to non-degeneracy, we conclude that there is some agent i who only receives full datasets, i.e., $x_{i,j} = \{0, 1\}$ for all j .

For such an agent i , budget exhaustion implies that $b_i = \sum_{j: x_{i,j}=1} p_j$. Since each $p_j = r_j \cdot p_{j_0}$, we have $b_i = p_{j_0} \cdot \sum_{j: x_{i,j}=1} r_j$. Since $b_i \in \mathbb{Q}^+$ and $r_j \in \mathbb{Q}^+$, we conclude that $p_{j_0} \in \mathbb{Q}^+$. This also implies that $p_j \in \mathbb{Q}^+$ for all j in T . Repeating the same argument for every component in G_p proves the theorem. $\square$

C.2 Multiplicity and non-convexity of CE

THEOREM 3.3. *There exists a data marketplace instance where there are multiple CE. Further, the set of CE is not convex.*

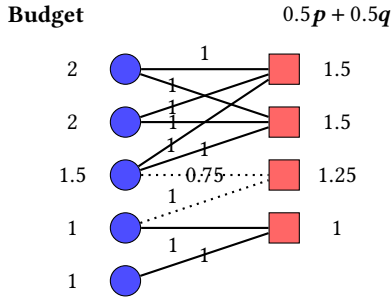


Figure 2: Example showing non-convexity of CE in the proof of Theorem 3.3.

PROOF. Consider a data marketplace with five buyers and four sellers. The budgets and two parameters are illustrated in Fig. 3. We provide two price vectors p and q , each of which can correspond to a CE along with some allocation. All datasets have a supply of 1. In the first price vector p , all the prices are set to 1. Hence, for buyer 3, the bang-for-the-buck of dataset $\mathcal{D}_3$ is $0.75/1 = 0.75$, which is smaller than those of the first two datasets. Meanwhile, for buyer 4, the last two datasets are the same. Thus, p forms a CE with the following allocation:

$$x_{1,1} = x_{1,2} = x_{2,1} = x_{2,2} = 1, x_{3,1} = 0.5, x_{3,2} = 1, x_{4,3} = x_{5,4} = 1$$

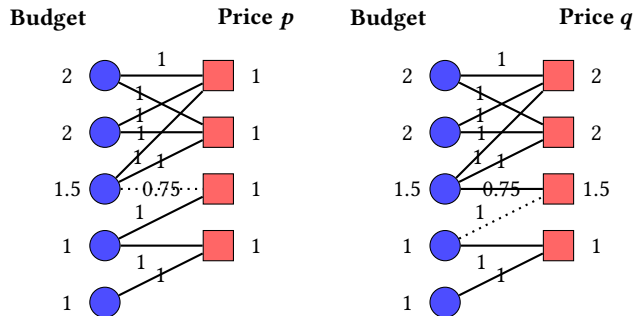


Figure 3: Two price vectors for the same data marketplace, where the square nodes represent buyers and the rounded nodes represent sellers. The solid edges represent MBB edges while the dotted edges represent non-MBB edges. For every seller i and buyer j with no edge between, $\tau_{i,j}$ is zero.

Meanwhile, as the prices of the first two datasets are doubled while the price of the third dataset increases by only a factor of 0.5. The dataset $\mathcal{D}_3$ becomes the MBB dataset for buyer 3. Therefore, the following allocation also forms a CE at the price vector q :

$$x_{1,1} = x_{2,2} = 1, x_{3,3} = 1, x_{4,4} = x_{5,4} = 1$$

We can observe that the budget of every buyer is totally spent on the MBB datasets, and the market-clearing condition is also satisfied. Next, consider the convex combination of the two price vectors, $w = 0.5p + 0.5q$, as illustrated in Fig. 2.

The price of $\mathcal{D}_3$ is then set as 1.25. Hence, its bang-for-the-buck for buyer 3 is $0.75/1.25 = 0.6$, which is smaller than those of the first two datasets, $1/1.5 = 0.667$. Meanwhile, the bang-for-the-buck of the fourth dataset for buyer 4 is $1/1 = 1$, which is also larger than that of $\mathcal{D}_3$. Therefore, no MBB edge is incident to dataset $\mathcal{D}_3$, which means that in an allocation that forms a CE with w , dataset $\mathcal{D}_3$ should not be allocated to any buyer, which contradicts the market-clearing condition. $\square$

D Omitted Proofs of Section 4

THEOREM D.1. *algorithm 1 computes an ε -CE for an ε -perturbed instance in $O\left(\frac{m}{\varepsilon} \cdot \log\left(\frac{2nb_{\max}\tau_{\max}}{b_{\min}\tau_{\min}}\right)\right)$ iterations.*

PROOF. Since the initial prices and allocation satisfy Invariant 1 (Lemma 4.4), and the allocation and price update steps maintain it (Lemma 4.5), we conclude by induction that Algorithm 1 maintains Invariant 1 throughout its execution. Thus, if Algorithm 1 terminates with prices p and allocation x , then (p, x) constitutes an ε -CE. This is because Invariant 1 ensures the MBB and market-clearing conditions are satisfied, and approximate budget exhaustion holds at termination since Algorithm 1 continues until the surplus $\Delta_i(p, x)$ of every buyer i is at most $\varepsilon \cdot b_i$. Therefore, it suffices to prove that Algorithm 1 terminates.

For this, we use the potential function $\phi(p) = \sum_j \log p_j$. Observe that the price of any dataset does not decrease during the algorithm. Moreover, at least one of the prices is increased by a factor of $(1 - \varepsilon)^{-1}$ at every price update step. Since the allocation update step does not change prices, we obtain that the potential function increases additively by $-\log(1 - \varepsilon)$ in every iteration. The proof of termination and the bound on time complexity follows once we establish that the potential function is bounded by above throughout the run of the algorithm, and a lower bound on its initial value $\phi(p^0)$.

For the upper bound, observe that:

$$\begin{aligned} \sum_{j \in M} p_j &= \sum_{j \in M} p_j \max_{i \in N} x_{i,j} && \text{(market-clearing)} \\ &\leq \sum_{j \in M} p_j \sum_{i \in N} x_{i,j} \\ &= \sum_{i \in N} \sum_{j \in M} p_j \cdot x_{i,j} \\ &\leq \sum_{i \in N} b_i && \text{(Property (iv) of Invariant 1)} \end{aligned}$$

Using the above inequality with the concavity of $\log(\cdot)$, we obtain the following upper bound:

$$\phi(p) = \sum_{j \in M} \log p_j \leq m \log \left(\frac{\sum_{j \in M} p_j}{m} \right)$$

$$\begin{aligned}
&\leq m \log \left(\frac{\sum_{i \in N} b_i}{m} \right) \\
&\leq m \log \left(\frac{nb_{\max}}{m} \right). \tag{3}
\end{aligned}$$

Next, we provide a lower bound on the value of the potential function $\phi(\mathbf{p}^0)$ using Lemma 4.4:

$$\phi(\mathbf{p}^0) = \sum_{j \in M} \log p_j^0 \geq m \log \left(\frac{(1-\varepsilon)\tau_{\min}b_{\min}}{m\tau_{\max}} \right). \tag{4}$$

Since every iteration increases the potential function additively by $-\log(1-\varepsilon)$, the bounds Eq. (3) and Eq. (4) imply that the number of iterations is at most

$$\frac{\phi(\mathbf{p}) - \phi(\mathbf{p}^0)}{-\log(1-\varepsilon)} \leq \frac{m}{-\log(1-\varepsilon)} \cdot \left(\log \left(\frac{nb_{\max}}{m} \right) - \log \left(\frac{(1-\varepsilon)\tau_{\min}b_{\min}}{m\tau_{\max}} \right) \right)$$

$$\leq \frac{m}{\varepsilon} \cdot \log \left(\frac{2nb_{\max}\tau_{\max}}{b_{\min}\tau_{\min}} \right),$$

where we used $-\log(1-\varepsilon) \geq \varepsilon$ and $\varepsilon < 1/2$. This proves the theorem. $\square$

Finally, we note that Theorem 4.1 follows from Theorem D.1 and Lemma 4.3.

REMARK 2. We remark that the existence of *exact* competitive equilibrium follows from the existence of ε -CE together with compactness arguments. In more detail, we note that the set of ε -CE is compact for any $\varepsilon \in (0, 1)$. We then consider the sequence $\{(\mathbf{p}^k, \mathbf{x}^k)\}_{k \geq 1}$, where $(\mathbf{p}^k, \mathbf{x}^k)$ is a $\frac{1}{k}$ -CE. By the Bolzano-Weierstrass Theorem, there exists a convergent subsequence that converges to $(\mathbf{p}^*, \mathbf{x}^*)$, since the set of approximate-CE is compact. We can then argue that $(\mathbf{p}^*, \mathbf{x}^*)$ must be an exact-CE.